

Clifford Algebras in Theoretical Physics

Seminar Methodology, Modular Syllabus, and Thematic Progression

Clifford Seminar Group

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1 Overview

This seminar investigates invariant geometric formulations of theoretical physics through Clifford algebras. The Kepler problem and the associated Coulomb potential serve as the central dynamical backbone and comparative benchmark across classical mechanics, spinor regularization, relativistic spacetime dynamics, and relativistic quantum mechanics. While this central force problem guides the thematic progression, the curriculum is designed to accommodate systematic deviations into adjacent physical and mathematical domains, including coordinate-free differential forms, monogenic analysis, gauge theories of gravitation, and geometric thermodynamics.

The colloquium operates as a collaborative working seminar focused on direct blackboard derivations and critical appraisal of physical foundations. The syllabus is structured into sequential modular topics. A given topic may span one or multiple 90-minute sessions depending on the analytical depth of the derivations and the scope of plenary debate. This modular design preserves structural continuity when topics expand or when tangential inquiries are pursued.

Exposition duties are distributed across senior–junior tandems to foster rigorous peer mentorship.

2 Methodology

2.1 Working Roles

The seminar must avoid passive lecture formats. Each session relies on four distinct responsibilities.

Referat (Lead Exposition) Delivered jointly by the assigned tandem. The presentation focuses on direct, blackboard derivations of primary equations. Conceptual assumptions and algebraic transitions take precedence over projected slides.

Korreferat (Respondent Critique) An assigned participant presents a critical commentary immediately following the Referat. The respondent examines implicit hypotheses, compares the results with traditional coordinate or matrix formalisms, identifies potential pathologies, and formulates targeted questions for discussion.

Thesenpapier (Position Paper) A concise memorandum circulated by the presenting tandem at least five days prior to the session. It contains formal definitions, key algebraic identity statements, and explicit theses concerning physical interpretation. Participants study this document in advance.

Sitzungsprotokoll (Session Record) A rotating participant documents the derivation steps, the resolution of theoretical debates, and open problems. The finalized protocol is archived in the shared repository within seven days following the session.

2.2 Tandem Collaboration

Beginning with Topic 2, sessions pair an advanced participant (senior) with an earlier-stage student (junior). The senior ensures mathematical coherence and contextualizes advanced algebraic techniques. The junior works directly through the derivation machinery, receiving structured guidance in multilinear structures and relativistic dynamics. Responsibilities for blackboard presentation are shared equally between both speakers.

2.3 Session Structure

Preparatory meetings with the seminar leader occur two weeks and five days prior to each session. Every attendee must submit at least one written question on the Thesenpapier 24 hours before meeting. The 90-minute session adheres to the allocation in table 1.

Table 1: Allocation of time per 90-minute seminar session.

Phase	Duration	Content	Responsible
I	40 min	Live blackboard derivation (Referat)	Presenting tandem
II	10 min	Critical response and opening theses (Korreferat)	Respondent
III	35 min	Plenary debate and interpretation	All participants
IV	5 min	Synthesis of open questions and preview	Seminar leader

3 Schedule and Syllabus

Primary reference works are designated by short abbreviations. VR indicates Vaz and da Rocha, *An Introduction to Clifford Algebras and Spinors* [1]; DL indicates Doran and Lasenby, *Geometric Algebra for Physicists* [2]; HS indicates Hestenes and Sobczyk, *Clifford Algebra to Geometric Calculus* [3]. Specific chapters and foundational literature are cited directly in table 2.

Table 2: Modular seminar syllabus and thematic sequence.

Topic	Subject	Mathematical and Physical Content	Literature
<i>Part A: Foundations and Geometric Analysis</i>			
1	Dual spaces, multilinear forms, and quadratic spaces	Dual linear space V^* , dual pairing $\langle \cdot, \cdot \rangle$, tensor algebra $T(V)$, symmetric bilinear forms, quadratic forms $q(v)$, polarization, non-degeneracy, and signature (p, q) over $\mathbb{R}$.	VR [1, Chap. 1]; Axler [4, Chaps. 1–2]; Fraleigh [5, Chaps. 1.1–1.2, 7.33].
2	Universal Clifford construction and involutions	Ideal $\mathcal{I}_q = \langle v \otimes v - q(v)1 \rangle$, quotient algebra $\mathcal{Cl}(V, q)$, universal property, $\mathbb{Z}_2$ -grading, grade involution $\hat{u}$, reversion $\tilde{u}$, Clifford conjugation $\bar{u}$, geometric product $uv = u \cdot v + u \wedge v$.	
3	Multivector structures, Lie algebras, and Spin groups	Module isomorphism $\mathcal{Cl}(V, q) \cong \bigwedge V$, blade projections, bivector Lie algebra $\mathcal{Cl}^2(V, q) \cong \mathfrak{so}(p, q)$, Clifford and Lipschitz groups, $\text{Pin}(p, q)$, $\text{Spin}(p, q)$, and rotor transformations $v \mapsto RvR$.	

Table 2: Modular seminar syllabus and thematic sequence.

Topic	Subject	Mathematical and Physical Content	Literature
4	Clifford analysis and monogenic fields	Vector derivative ∇ , Cauchy–Riemann generalized operator, monogenic multivector fields $\nabla\Phi = 0$, Cauchy kernel, Borel–Pompeiu integral formula, and coordinate-free multipole expansions of extended Keplerian sources.	
<i>Part B: Classical Kepler Dynamics in $\mathcal{Cl}(3, 0)$</i>			
5	Central forces and orbital angular momentum	Equation of motion $\dot{p} = -(k/r^3)r$, orbital angular momentum bivector $L = r \wedge p$, conservation proof $\dot{L} = 0$, planar confinement $r \wedge L = 0$, coordinate-free duality via volume pseudoscalar I_3 .	
6	Laplace–Runge–Lenz vector and conics	Multivector construction $A = \frac{1}{mk}pL - \hat{r}$, algebraic proof of $\dot{A} = 0$ without coordinate frames, orbit derivation via scalar contraction $r \cdot A$, and invariant classification of conics.	
<i>Part C: Regularization, Harmonic Analysis, and Hidden Symmetries</i>			
7	Kustaanheimo–Stiefel transformation	Regularization of collision singularities at $r \rightarrow 0$, spinor mapping $x = u\sigma_3\tilde{u}$ with $u \in \mathcal{Cl}^+(3, 0) \cong \mathbb{H}$, Sundman time rescaling $dt = r ds$, and $U(1)$ gauge freedom.	
8	Four-dimensional oscillator and $\text{SO}(4)$	Linearization to $u'' + \frac{ E }{2}u = 0$, four-dimensional isotropic harmonic oscillator, realization of $\mathfrak{so}(4) \cong \mathfrak{su}(2) \oplus \mathfrak{su}(2)$ dynamical symmetry via L and A , bound-state degeneracy.	
9	Clifford–Fourier analysis and Fock projection	Clifford–Fourier transform with pseudoscalar kernel $e^{-I_3 k \cdot x}$, multivector uncertainty relations, Fock stereographic projection of the Coulomb problem onto S^3 , and bound states as spherical monogenics.	

Part D: Relativistic Dynamics in Spacetime Algebra $\mathcal{Cl}(1, 3)$

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Topic	Subject	Mathematical and Physical Content	Literature
10	Spacetime algebra and observer split	Dirac generators $\{\gamma_\mu\}$, signature $(+, -, -, -)$, observer split via γ_0 , relative spatial bivectors $\sigma_k = \gamma_k \gamma_0$, proper Lorentz rotors in $\text{Spin}^+(1, 3)$, rotor kinematics $\dot{R} = \frac{1}{2}\Omega R$, Thomas precession.	
11	Relativistic electrodynamics and Lorentz force	Spacetime vector derivative $\nabla = \gamma^\mu \partial_\mu$, Faraday bivector $F = E + I_4 B$, Maxwell system $\nabla F = J$, relativistic equation of motion $\dot{p} = eF \cdot u$, proper-time rotor integration.	
12	Relativistic central forces and perihelion advance	Relativistic central potentials, breakdown of LRL vector conservation, energy-dependent dynamical corrections, and calculation of special-relativistic perihelion advance.	
13	Gauge Theory Gravity and planetary orbits	Position gauge field $\bar{h}(a)$ and rotation gauge field $\bar{\Omega}(a)$ on Minkowski background, spherically symmetric solutions, geodesic rotor trajectories, and general-relativistic orbital precession.	
14	Geometric thermodynamics in spacetime algebra	Conserved particle flux $N = nu$, stress-energy operator $T(a)$, entropy current $S = su$, invariant thermal 1-vector $\beta = u/(k_B T)$, Killing equilibrium $\nabla \wedge \beta = 0$, and Tolman–Ehrenfest thermal gradients in static gravitational fields.	

Part E: Quantum Kepler Problem and Dirac–Hestenes Theory

15	Real Pauli algebra and spatial spin	Spatial algebra $\mathcal{Cl}(3, 0)$ without matrix representations, unit spatial vectors as spin observables, Pauli multivector wave equation, spin-orbit coupling from the geometric product.	
16	Dirac–Hestenes field equation	Spacetime algebra representation of the Dirac equation, even multivector field $\Psi \in \mathcal{Cl}^+(1, 3)$, spin plane $\gamma_2 \gamma_1$, polar form $\Psi = \rho^{1/2} e^{I_4 \beta/2} R$, probability current $J = \Psi \gamma_0 \tilde{\Psi}$, spin bivector $S = \frac{1}{2} \hbar \Psi \gamma_3 \tilde{\Psi}$.	

Table 2: Modular seminar syllabus and thematic sequence.

Topic	Subject	Mathematical and Physical Content	Literature
17	Exact solution of the relativistic hydrogen atom	Separation of variables in spherical coordinates within $\mathcal{Cl}^+(1, 3)$, rotor spherical harmonics, radial differential equations in the even sub-algebra, derivation of the Sommerfeld–Dirac energy spectrum.	
18	Fine structure, Darwin term, and zitterbewegung	Non-relativistic expansion of the Dirac–Hestenes equation, geometric origin of the Darwin correction, rotor phase dynamics, lightlike field circulation, field-theoretic interpretation of zitterbewegung.	

4 Tandem Assignments

The assignments for lead expositions, junior collaborators, and designated respondents are compiled in table 3.

References

- [1] Jayme Vaz and Roldão da Rocha. *Introduction to Clifford Algebras and Spinors*. Oxford: Oxford university press, 2016. 128 pp. ISBN: 978-0-19-878292-6.
- [2] Chris Doran and Anthony Lasenby. *Geometric Algebra for Physicists*. Cambridge: Cambridge University Press, 2013. 578 pp. ISBN: 978-0-521-48022-2. DOI: [10.1017/CB09780511807497](https://doi.org/10.1017/CB09780511807497).
- [3] David Hestenes and Garret Sobczyk. *Clifford Algebra to Geometric Calculus*. Dordrecht: Springer Netherlands, 1984. ISBN: 978-90-277-2561-5. DOI: [10.1007/978-94-009-6292-7](https://doi.org/10.1007/978-94-009-6292-7). URL: <http://link.springer.com/10.1007/978-94-009-6292-7>.
- [4] Sheldon Axler. *Linear Algebra Done Right*. Undergraduate Texts in Mathematics. Cham: Springer International Publishing, 2015. ISBN: 978-3-319-11079-0. DOI: [10.1007/978-3-319-11080-6](https://doi.org/10.1007/978-3-319-11080-6). URL: <https://link.springer.com/10.1007/978-3-319-11080-6>.
- [5] John B. Fraleigh and Victor J. Katz. *A First Course in Abstract Algebra*. 8th ed. World Student Series. Hoboken, New Jersey: Pearson, 2021. 424 pp. ISBN: 978-0-13-575816-8.

Table 3: Register of seminar leaders, tandems, and respondents across topics.

Topic	Subject	Senior / Lead	Junior	Respondent
1	Dual spaces and quadratic forms	J. L. Avila-Martinez	—	
2	Universal Clifford construction	J. L. Avila-Martinez	C. Martinez	
3	Multivector structures and Spin			
4	Clifford analysis and monogenics			
5	Central forces and bivectors			
6	LRL vector and conics			
7	KS transformation			
8	4D oscillator and $SO(4)$			
9	Clifford–Fourier and Fock projection			
10	Spacetime algebra			
11	Electrodynamics in STA			
12	Relativistic central forces			
13	Gauge Theory Gravity			
14	Geometric thermodynamics			
15	Pauli algebra and spin			
16	Dirac–Hestenes equation			
17	Relativistic hydrogen			
18	Fine structure and zitterbewegung			